

Open book exam. Submit your answers to gradescope.

In your explanations and justifications, write complete and grammatically correct sentences.

Please do not ask questions during the exam.

4. Let $G = (V, E)$ be a weighted graph with $V = \{0, 1, 2, 3, 4, 5\}$ and edges $E = \{((0, 2), 9), ((0, 1), 9), ((0, 4), 9), ((1, 4), 6), ((2, 5), 8), ((1, 2), 7), ((2, 3), 8), ((3, 5), 2), ((4, 5), 3), ((0, 5), 9)\}$, where edge (i, j) with weight w is represented by $((i, j), w)$.
- Construct a minimum spanning tree with Kruskal's algorithm.
Show all steps in the evolution of the union-find data structure.
 - Explain how definition of the merge operation is important for the $O(m \log(m))$ cost for a graph with m edges.

/15

Answer:

4. (a) U is the union-find data structure, T is the tree
- $U = [1, 2, 3, 4, 5, 4], T = [((3, 5), 2)]$
 - $U = [1, 2, 3, 4, 4, 4], T = [((3, 5), 2), ((4, 5), 3)]$
 - $U = [1, 4, 3, 4, 4, 4] T = [((3, 5), 2), ((4, 5), 3), ((1, 4), 6)]$
 - $U = [1, 4, 4, 4, 4, 4], T = [((3, 5), 2), ((4, 5), 3), ((1, 4), 6), ((1, 2), 7)]$
 - skipping $(2, 5)$
 - skipping $(2, 3)$
 - $U = [4, 4, 4, 4, 4, 4], T = [((3, 5), 2), ((4, 5), 3), ((1, 4), 6), ((1, 2), 7), ((0, 2), 9)]$
- (b) The merge operation is defined such that, when merging two components, the label of the smallest component is adjusted. Because of this, any sequence of k merge operations takes at most $O(k \log(k))$ time. In a graph with n vertices, as at most $n - 1$ merge operations are done, the cost of all merge operations is $O(n \log(n))$, which is bounded by $O(m \log(m))$ for a connected graph.